

Question No : 1 of 33

Marks: 1 (Time 1 Min)



Curl of a vector field is linked with

00:44

Answer

- ☐ Greatest rate of increase in function
- ☒ volume density of the outward flux of a vector field
- ☐ Rotational properties of field





A volume integral refers to an integral over a

00:55

Answer

☐ 1-dimensional domain

☐ 2-dimensional domain

☒ 3-dinemsional domain



☐ None of these



Volume integral in Spherical coordinates can be expressed as

01:04

Answer



$$\iint_{\mathbf{R}} g(\rho, \varphi, z) d\rho d\varphi$$



$$\iiint_{\mathbf{R}} g(\rho, \varphi, z) d\rho d\varphi dz$$



$$\iiint_{\mathbf{R}} r \sin \varphi g(\rho, \varphi, z) d\rho d\varphi dz$$

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The divergence theorem converts

Answer



Line to surface integral



Surface to volume integral



Volume to line integral



Surface to line integral

01:16

The triple integral is used to compute volume.

Answer

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True



False



The Stoke's theorem uses which of the following operation?

01:30

Answer

4 / 17



Divergence



Gradient



Curl



Laplacian

The Stoke's theorem can be used to find which of the following?

Answer

- ☐ 01:41 Area enclosed by a function in the given region ✓
- ☒ Volume enclosed by a function in the given region ✗
- ☐ Linear distance
- ☐ Curl of the function

The Stoke's theorem can be used to find which of the following?

Answer



Area enclosed by a function in the given region

01:46



~~Volume~~ enclosed by a function in the given region



Linear distance



Curl of the function



Applications of Green's theorem are meant to be in

01:51

Answer

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One dimensional



Two dimensional



Three dimensional



Four dimensional

The Green's theorem can be related to which of the following theorems mathematically?

Answer



Gauss divergence theorem

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Stoke's theorem



Euler's theorem



Leibnitz's theorem



Which of the following is not an application of Green's theorem?

02:10

Answer



Solving two dimensional flow integrals



Area surveying



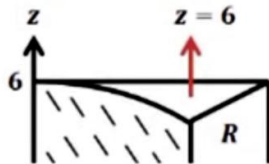
Volume of plane figures



Centroid of plane figures



The surface S and its projection R on the yz -plane are shown in figure.



Answer



$y=0$ to $y=4$ and $z=0$ to $z=2$

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$y=0$ to $y=4$ and $z=0$ to $z=6$



$y=0$ to $y=2$ and $z=0$ to $z=4$



None of these

02:17



The Laplace operator is _____ order differential operator given by the divergence of the gradient of a given function defined over a space R .

02:39

Answer



first



second



third



All of above

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Marks: 1 (Time 1 Min)

02:48

An integral where the function is evaluated along a _____ is called line integral.

Answer

☐ line

☒ curve

☐ scalar point



A vector is said to be solenoidal when its _____.

Answer



divergence is zero



divergence is unity

02:54



curl is zero



curl is unity

The Green's theorem in plane is a special case of _____ theorem.

Answer



Divergence

03:02



Gauss divergence

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stokes



line integral





The generalization of Green's theorem as Gauss's divergence theorem is also called the _____ vector form of Green's theorem.

Answer

First



Second



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Third



Fourth

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$T = \frac{dr}{ds}$ is the vector to the curve C

Answer



unit tangent



unit normal



position



None of the given choices

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Which of the following is perpendicular to the given surface $\phi(x,y,z) = c$, where c is a constant?

Answer



$\nabla(2\phi)$



$\nabla\phi$



Both $\nabla\phi$ and $\nabla(2\phi)$



Neither $\nabla\phi$ nor $\nabla(2\phi)$



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For a given vector $\vec{V}$, which of the following is a scalar quantity?

Answer

☐ $2\vec{V}$

☐ $\vec{V} \times \vec{V}$

☐ $\nabla \times \vec{V}$

☒ $\nabla \cdot \vec{V}$

03:56

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The function $\varphi(x,y,z) = x + y + z^2$ is a

Answer

☐ Vector Function

☒ Scalar function

☐ Vector Field

☐ None of these

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Type here to search



Question No : 21 of 33

Marks: 1 (Time 1 Min)



$$\nabla \times (\varphi \vec{A}) = ?$$

04:24

Answer



$$\varphi(\nabla \times \vec{A}) + \vec{A} \times (\varphi \times \nabla)$$



$$\varphi(\nabla \times \vec{A}) + (\varphi \nabla) \times \vec{A}$$



$$(\varphi \times \vec{A})\nabla + \vec{A} \times (\varphi \nabla)$$

Let R be the projection of the surface S on the xy - plane, then $\iint_S \vec{A} \cdot \hat{n} dS = ?$

04:41

Answer

☐ $\iint_R \vec{A} \hat{n} dx dy$

☒ $\iint_R \vec{A} \hat{n} \frac{dx dy}{|\hat{n} \cdot \vec{k}|}$

L

☐ $\iint_R \vec{A} \hat{n} \frac{dx dy}{|\vec{A} \cdot \vec{k}|}$

☐ None of these

If R is the region bounded by a closed surface S and $\vec{A}$ is a vector point function

with continuous first partial derivative then $\iint_S \vec{A} \cdot \hat{n} \, dS =$

Answer

☐ $\iiint_R \nabla \cdot \hat{n} \, dS$

☐ $\iiint_R \nabla \cdot \hat{n} \, dV$

☒ $\iiint_R \nabla \cdot \vec{A} \, dS$



If R is the region bounded by a closed surface S and $\vec{A}$ is a vector point function

with continuous first partial derivative then $\iint_S \vec{A} \cdot \hat{n} \, dS =$

Answer

☐ $\iiint_R \nabla \cdot \hat{n} \, dS$

☐ $\iiint_R \nabla \cdot \hat{n} \, dV$

☒ $\iiint_R \nabla \cdot \vec{A} \, dS$

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☐ $\iiint_R \nabla \cdot \vec{A} \, dV$



According to Stokes' theorem if S is an open, two sided surface bounded by a simple closed

curve C , and $\vec{A}$ has continuous first partial derivatives then $\oint_C \vec{A} \cdot d\vec{r} = ?$

05:47

Answer

☐ $\int_S \nabla \cdot \vec{A} \, dS$

☐ $\int_S \nabla \times \vec{A} \, dS$

☒ $\int (\nabla \times \vec{A}) \cdot \hat{n} \, dS$



According to stock's theorem, if $\int_C \vec{A} \cdot d\vec{r} = 0$, for every closed curve C is _____ = 0 identically.

06:14

Answer



$$\nabla \cdot \vec{A}$$



$$\nabla \times \vec{A}$$



$$\nabla \cdot d\vec{r}$$

The normal vector to the surface $f(x, y, z) = 0$ is $\vec{N} = \underline{\hspace{2cm}}$.

Answer



$\nabla \times f(x, y, z)$

X



$\nabla \cdot f(x, y, z)$

06:26



$\nabla f(x, y, z)$





Question No : 28 of 33

Marks: 3 (Time 6 M)

Check if the following integral is dependent or independent of path $\int_{(0,0)}^{(2,1)} (3x^2 - 2xy^2) dx - 2x^2 y dy$

06:36

Answer



Styles



Format



Font



Size

A**A**

Question No : 29 of 33

Marks: 3 (Time 61)

Find the limits of integration for $\iiint_V \vec{F} dV$ where V is the region bounded by the surfaces $x=0, y=0, z=0$ and $x+y+z=4$

Answer



Styles

Format

Font

Size

A.

A



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$$\frac{1}{2} \oint_C x dy - y dx$$

Answer

07:20



07:38

Find a unit normal to the surface $x^2y - 3z = 2$ at the point $(1,1,1)$.

Answer